

Data Science Learning Material

1. In an A/B test, how can you check if assignment to the various buckets was truly random?

Plot the distributions of multiple features for both A and B and make sure that they have the same shape. More rigorously, we can conduct a permutation test to see if the distributions are the same.

MANOVA to compare different means

2. What would be the hazards of letting users sneak a peek at the other bucket in an A/B test?

The user might not act the same suppose had they not seen the other bucket. You are essentially adding additional variables of whether the user peeked the other bucket, which are not random across groups.

3. How would you run an A/B test for many variants, say 20 or more?

one control, 20 treatment, if the sample size for each group is big enough.

Ways to attempt to correct for this include changing your confidence level (e.g. Bonferroni Correction) or doing family-wide tests before you dive in to the individual metrics (e.g. Fisher's Protected LSD).

4. How would you run an A/B test if the observations are extremely right-skewed?

lower the variability by modifying the KPI

cap values

percentile metrics

log transform

5. I have two different experiments that both change the sign-up button to my website. I want to test them at the same time. What kinds of things should I keep in mind?

exclusive -> ok

6. What is a p-value? What is the difference between type-1 and type-2 error?

A p-value is defined such that under the null hypothesis less than the fraction p of events have parameter values more extreme than the observed parameter. It is not the probability that the null hypothesis is wrong.

type-1 error: rejecting H_0 when H_0 is true

type-2 error: not rejecting H_0 when H_a is true

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7. You are AirBnB and you want to test the hypothesis that a greater number of photographs increases the chances that a buyer selects the listing. How would you test this hypothesis?

For randomly selected listings with more than 1 pictures, hide 1 random picture for group A, and show all for group B. Compare the booking rate for the two groups.

8. How would you design an experiment to determine the impact of latency on user engagement?

The best way I know to quantify the impact of performance is to isolate just that factor using a slowdown experiment, i.e., add a delay in an A/B test.

9. What is maximum likelihood estimation? Could there be any case where it doesn't exist?

A method for parameter optimization (fitting a model). We choose parameters so as to maximize the likelihood function (how likely the outcome would happen given the current data and our model).

maximum likelihood estimation (MLE) is a method of estimating the parameters of a statistical model given observations, by finding the parameter values that maximize the likelihood of making the observations given the parameters. MLE can be seen as a special case of the maximum a posteriori estimation (MAP) that assumes a uniform prior distribution of the parameters, or as a variant of the MAP that ignores the prior and which therefore is unregularized.

for gaussian mixtures, non parametric models, it doesn't exist

10. What's the difference between a MAP, MOM, MLE estimator? In which cases would you want to use each?

MAP estimates the posterior distribution given the prior distribution and data which maximizes the likelihood function. MLE is a special case of MAP where the prior is uninformative uniform distribution.

MOM sets moment values and solves for the parameters. MOM is not used much anymore because maximum likelihood estimators have higher probability of being close to the quantities to be estimated and are more often unbiased.

11. What is a confidence interval and how do you interpret it?

For example, 95% confidence interval is an interval that when constructed for a set of samples each sampled in the same way, the constructed intervals include the true mean 95% of the time.

if confidence intervals are constructed using a given confidence level in an infinite number of independent experiments, the proportion of those intervals that contain the true value of the parameter will match the confidence level.

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12. What is unbiasedness as a property of an estimator? Is this always a desirable property when performing inference? What about in data analysis or predictive modeling?

Unbiasedness means that the expectation of the estimator is equal to the population value we are estimating. This is desirable in inference because the goal is to explain the dataset as accurately as possible. However, this is not always desirable for data analysis or predictive modeling as there is the bias variance tradeoff. We sometimes want to prioritize the generalizability and avoid overfitting by reducing variance and thus increasing bias.

13. What is Selection Bias?

Selection bias is a kind of error that occurs when the researcher decides who is going to be studied. It is usually associated with research where the selection of participants isn't random. It is sometimes referred to as the selection effect. It is the distortion of statistical analysis, resulting from the method of collecting samples. If the selection bias is not taken into account, then some conclusions of the study may not be accurate.

- The types of selection bias include:
- Sampling bias: It is a systematic error due to a non-random sample of a population causing some members of the population to be less likely to be included than others resulting in a biased sample.
- Time Interval bias: A trial may be terminated early at an extreme value (often for ethical reasons), but the extreme value is likely to be reached by the variable with the largest variance, even if all variables have a similar mean.
- Data: When specific subsets of data are chosen to support a conclusion or rejection of bad data on arbitrary grounds, instead of according to previously stated or generally agreed criteria.
- Attrition: Attrition bias is a kind of selection bias caused by attrition (loss of participants) discounting trial subjects/tests that did not run to completion.

14. Probability Basics and Random Variables

The beginnings of probability start with thinking about sample spaces, basic counting and combinatorial principles. Although it is not necessary to know all of the ins-and-outs of combinatorics, it is helpful to understand the basics for simplifying problems. One classic example here is the “stars and bars” counting method.

The other core topic to study is random variables. Knowing concepts related to expectation, variance, covariance, along with the basic probability distributions is crucial.

15. Probability Distributions

For modeling random variables, knowing the basics of various probability distributions is essential. Understanding both discrete and continuous examples, combined with expectations and variances, is crucial. The most common distributions discussed in interviews are the Uniform and Normal but there are plenty of other well-known distributions for particular use cases (Poisson, Binomial, Geometric).

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Most of the time knowing the basics and their applications should suffice. For example, which distribution would flipping a coin be under? What about waiting for an event? It never hurts being able to do the derivations for expectation, variance, or other higher moments.

16. Hypothesis Testing

Hypothesis testing is the backbone behind statistical inference and can be broken down into a couple of topics. The first is the Central Limit Theorem, which plays an important role in studying large samples of data. Other core elements of hypothesis testing: sampling distributions, p-values, confidence intervals, type I and II errors. Lastly, it is worth looking at various tests involving proportions, and other hypothesis tests.

Most of these concepts play a crucial role in A/B testing, which is a commonly asked topic during interviews at consumer-tech companies like Facebook, Amazon, and Uber. It's useful to not only understand the technical details but also conceptually how A/B testing operates, what the assumptions are, possible pitfalls, and applications to real-life products.

17. Modeling

Modeling relies on a strong understanding of probability distributions and hypothesis testing. Since it is a broad term, we will refer to modeling as the areas which have a strong statistical intersection with Machine Learning. This includes topics such as: linear regression, maximum likelihood estimation, & bayesian statistics. For interviews focused on modeling and machine learning, knowing these topics is essential.

18. What is a *Probability Distribution*?

A **Probability Distribution** is a statistical function that describes all the possible values and likelihood that a random variable can take within a given range.

There are two main types of probability distribution:

- **Discrete probability distributions:** used for random variables with discrete outcomes, for example, the number of heads in five consecutive coin tosses, the number of rainy days in a given week, the number of goals scored by a player, and so on.
- **Continuous probability distributions:** used for random variables with continuous outcomes, for example, the height of male students, median house prices in San Francisco, claim amounts experienced by an insurance company, and so on.

19. Presidential Election Challenge: What is the probability that A or B wins the election?

In a presidential election, there are four candidates. Call them A, B, C, and D. Based on our polling analysis, we estimate that A has a 20% chance of winning the election, while B has a 40% chance of winning. What is the probability that A or B wins the election?

Answer

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We first notice the events that A wins, B wins, C wins, and D wins are mutually exclusive events since more than one of them cannot occur at the same time. For example, if A wins, then B cannot win. We know that the probability of the union of two disjoint events is the summation of individual probabilities. Therefore,

$$\begin{aligned}P(A \text{ wins or } B \text{ wins}) &= P(\{A \text{ wins}\} \cup \{B \text{ wins}\}) \\P(A \text{ wins or } B \text{ wins}) &= P(\{A \text{ wins}\} \cup \{B \text{ wins}\}) \\&= P(\{A \text{ wins}\}) + P(\{B \text{ wins}\}) \\P(A \text{ wins or } B \text{ wins}) &= P(\{A \text{ wins}\}) + P(\{B \text{ wins}\}) \\&= 0.2 + 0.4 = 0.6 \\P(A \text{ wins or } B \text{ wins}) &= 0.2 + 0.4 = 0.6\end{aligned}$$

20. Rolling a fair die event challenge

Let's suppose I roll a fair die. Let A be the event that an outcome is an odd number and let B be the event that the outcome is less than or equal to 3. What is the probability $P(A|B)$?

Answer

Given $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$, if we know B has occurred, the outcome must be among $\{1, 2, 3\}$. For A to also happen the outcome must be in $A \cap B = \{1, 3\}$. Since all die rolls are equally likely, then we calculate the conditional probability as:

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{2}{3}$$

$$|A \cap B| = 2$$

$$2$$

21. What is the difference between a *Combination* and a *Permutation*?

A **Combination** is the choice of r elements from a set of n elements *without replacement* and where *order does not matter*. Is most used to group data. For example, picking three team members from a group, picking two colors from a color brochure, etc. It is mathematically defined as:

$$C_{n,r} = \frac{n!}{r!(n-r)!}$$

□ $n!$
□ A **Permutation** is the choice of r elements from a set of n elements *without replacement* and where the *order matters*. Is used to list data, for example picking first, second and third place winners, picking two favorite colors -in order- from a color brochure, etc. It is mathematically defined as:

$$P_{n,r} = \frac{n!}{(n-r)!}$$

$$\bullet \quad n!$$

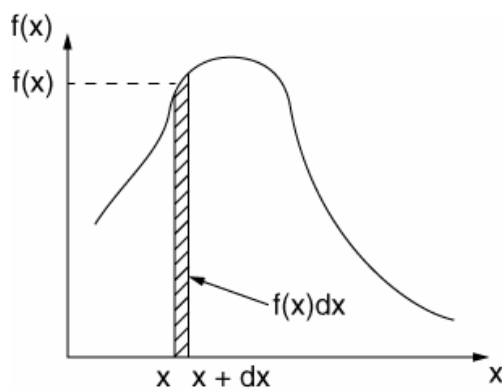
22. What is the difference between the *Bernoulli* and *Binomial* distribution?

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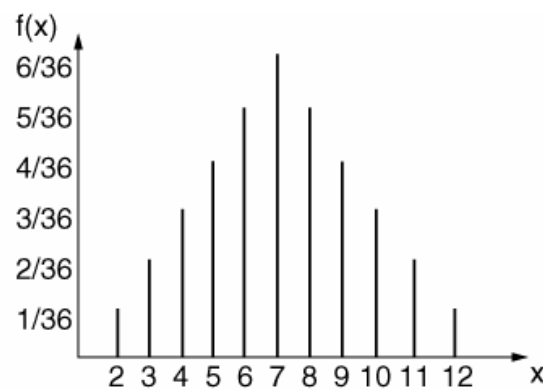
- The **Bernoulli distribution** is the *discrete probability distribution* of a random variable which takes a **binary** output: 1 with probability p , and 0 with probability $(1-p)$. The idea is that, whenever we are running an experiment that might lead either to *success* or to a *failure*, we can associate with our *success* (labeled with 1) a probability p , while our *failure* (labeled with 0) will have probability $(1-p)$.
- In the **Binomial distribution** we keep the same idea as before: we count *probability* of a *success* or *failure* outcome in an experiment, but this time it is *repeated multiple times*.

23. What's the difference between *Probability Mass Functions* and *Density Probability Functions*?

- **Probability mass functions** are used to describe *discrete probability* distributions and allow us to determine the probability of an observation being *exactly equal* to a target value.
- **Density functions** are used to describe *continuous probability* distributions and allows us to determine the probability of an observation being within a *range around our target value* by computing the *area under the curve* for our interval.



Continuous probability density function



Probability mass function

24. A coin was flipped 1000 times, and 550 times it showed up heads. Do you think the coin is biased?

To answer this question let's say XX is the number of heads and let's assume that the coin is not biased. Since each individual flip is a Bernoulli random variable, we can assume it has a probability of showing up heads as $p = 0.5$, so this will lead to the following expected number of heads:

$$\mu = np = 1000 \times 0.5 = 500$$

And the following standard deviation:

$$\sigma = np(1-p) = 1000 \times 0.5 \times 0.5 = 250 \approx 16$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{1000 \times 0.5 \times 0.5} = 250 \approx 16$$

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Given that we got a 1000 sample size, we can apply the Central Limit Theorem to approximate the total number of heads as normal distribution and calculate the corresponding z-score to test the hypothesis that the coin is fair.

$$z = \frac{x - \mu}{\sigma} = \frac{550 - 500}{16} = 3.125 > 3 = z_{\alpha}$$

$$x - \mu = 16$$

$$550 - 500 = 3.125 > 3$$

This means that, if the coin were fair, the event of seeing 550 heads should occur with a < 1% chance under normality assumptions. Therefore, the coin is likely biased.

25. Find a probability of dangerous Fire when there is Smoke

Let's say:

- the probability of dangerous fires are rare (1%)
- but smoke is fairly common (10%) due to barbecues,
- and 90% of dangerous fires make smoke

Can you find the probability of dangerous Fire when there is Smoke?

Answer

We can then discover the **probability of dangerous Fire when there is Smoke** using the Bayes' Theorem:

$$\begin{aligned} P(\text{Fire} \mid \text{Smoke}) &= P(\text{Fire}) * P(\text{Smoke} \mid \text{Fire}) / P(\text{Smoke}) \\ &= 0.01 * 0.9 / 0.1 \\ &= 0.09 \text{ (9\%)} \end{aligned}$$

So probability of dangerous fire when there is a smoke is 9%.

26. Flipping the Unfair Coin challenge

There is a *fair* coin (one side heads, one side tails) and an *unfair* coin (both sides tails). You pick one at random, flip it five times, and observe that it comes up as tails all five times. What is the chance that you are flipping the unfair coin?

Answer

Let the event of flipping the fair coin be denoted by F , the event of flipping the unfair coin be denoted by U and the event of tossing all 5 tails by T . We are required to find $P(U|T)$ and we can use the Bayes' theorem as:

$$P(U|T) = \frac{P(T|U)P(U)}{P(T|U)P(U) + P(T|F)P(F)}$$

$$P(T|U)P(U)$$

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where we use the law of total probability to compute $P(T)$ as:

$$P(T) = P(U)P(T|U) + P(F)P(T|F)$$

$P(T)$

Now, we know that:

- $P(T|U) = 1$ is the probability of the unfair coin flipping tails in a particular toss.
- $P(U) = P(F) = 0.5$ since we pick the unfair and the fair coin at random.
- $P(T|F)$ is the probability of the fair coin flipping five tails:

$$P(T|F) = \left(\frac{1}{2}\right)^5 = \frac{1}{32} \approx 0.031$$

$$P(T) \approx 0.031$$

Now, by putting all these values on the Bayes' rule formula we got:

$$P(U|T) = \frac{0.5 \cdot 1}{0.5 \cdot 1 + 0.5 \cdot 0.031} = 0.97$$

$$P(F|T) = \frac{0.5 \cdot 0.031}{0.5 \cdot 1 + 0.5 \cdot 0.031} = 0.031$$

Therefore the probability we picked the unfair coin is about 97%.

27. Given two fair dices, what is the probability that two dices sum to 8?

What is the probability that two dices sum to 8 when the first dice is 6?

Answer

Given the *probability* formula:

$$P = \frac{n}{n_{\text{total}}}$$

n

where n is the number of desired outcomes and n_{total} is the number of possible outcomes. For one single fair dice, we have 6 possible outcomes, so for two fair dices, we have $6 \cdot 6 = 36$ possible outcomes.

We want that the two dices sums 8 so the possible combinations are:

- (2,6), (3,5), (4,4)
- (5,3), (6,2)

We got that the number of desired outcomes are 5. The probability is then

$$P = \frac{5}{36} = 0.13888 \dots$$

$$P \approx 0.13888 \dots$$

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For the second part we use the *Bayes Theorem* and *conditional probability* formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B)$$

Let's say:

- A is the event of getting an 8
- B is the event of getting a 3.

Now, from the previous combination, we got that (3,5) is the only event that satisfies both A and B so then $P(A \cap B) = 1/36$. On other hand, the probability of getting a 3 in a fair dice is just $1/6$, we now put these values on the previous formula and compute:

$$P(A|B) = \frac{1/36}{1/6} = 1/6 = 0.1666 \dots$$

$$P(A \cap B) = 1/36$$

$$P(B) = 1/6$$

28. How do you transform a *Skewed Distribution* into a *Normal Distribution*?

To transform a **Skewed Distribution** into a **Normal Distribution** we apply some **linearized function** on it. Some common functions that achieve this goal are:

- **Logarithmic function:** We can use it to make *extremely* skewed distributions less skewed, especially for *right-skewed distributions*. The only condition is that this function is defined only for **strictly positive numbers**. $f(x) = \ln(x)$
- **Square root transformation:** this one has an average effect on distribution shape: it's weaker than *logarithmic transformation*, and it's also used for reducing *right-skewed distributions*, but is defined only for **positive numbers**. $f(x) = \sqrt{x}$

□ **Reciprocal transformation:** this one reverses the order among values of the same sign, so *large values* become *smaller*, but the *negative reciprocal* preserves the order among values of the same sign. The only condition is that this function is not defined for **zero values**.

$$f(x) = 1/x$$

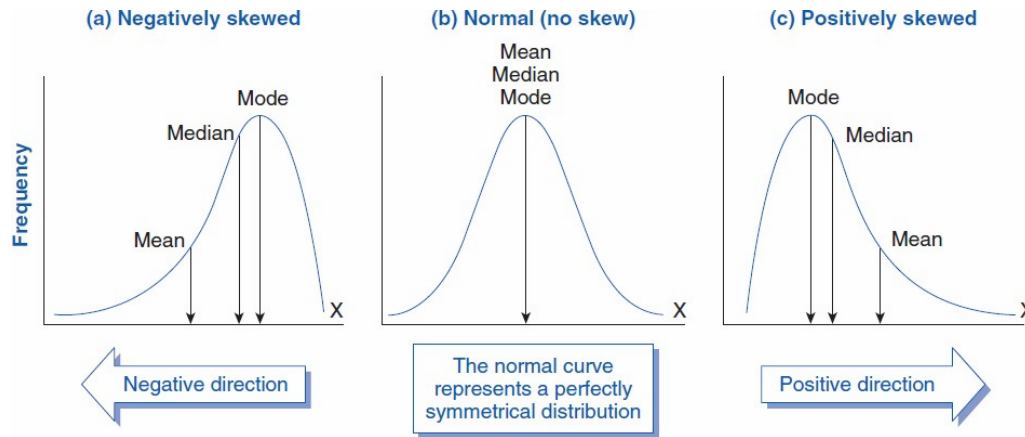
- **Exponential or Power transformation:** has a reasonable effect on distribution shape; generally, we apply power transformation (power of two usually) to reduce *left skewness*. We could also try any exponent to see which one provides better results. $f(x) = x^n$
- **Box-Cox Transformation:** in this transformation, we're searching and evaluating all the other transformations and *choosing the best one*. It's defined as:

$$xi(\lambda) = \begin{cases} xi\lambda - 1/\lambda & \text{if } \lambda \neq 0 \\ \ln xi & \text{if } \lambda = 0 \end{cases}$$

$$xi\lambda - 1/\lambda \text{ if } \lambda \neq 0, \ln xi \text{ if } \lambda = 0$$

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The exponent here is a variable called *lambda* (λ) that varies over the range of -5 to 5, and in the process of searching, we examine all values of λ . Finally, we choose the *optimal value* (resulting in the best approximation to a normal distribution) for the variable.



29. How would you test hypotheses using *Bayes' Rule*?

Bayes' Rule is an equation that allows us to calculate a *conditional probability* with new information we already have. Given two different events **A** and **B**, we can think in **Baye's theorem** in terms of two events:

- A hypothesis **A** which can be true or false.
- A evidence **B** which can be present or absent.

So in this way **A** is the event we want the probability of. Using the *Bayes' theorem* the problem can be set as:

$$P(A|B) = \frac{P(B|A)P(A)P(B)}{P(A|B)P(B)}$$

$$P(B|A)P(A)$$

where:

- **P(A|B)** is the probability of event **A** occurring given that event **B** is true. Is also called the *posterior* and is what are we trying to estimate.
- **P(B|A)** is the probability of event **B** occurring given that event **A** is true. It can also be interpreted as the *likelihood* and is the probability of observing the new evidence, given our initial hypothesis.
- **P(A)** and **P(B)** are the probabilities of observing **A** and **B** respectively without any given conditions; they are known as the *prior* and *marginal* probability, respectively.

Now, to test between two hypotheses we compute the probability of *each one* using the previous formula and compare them. The hypothesis with the *higher posterior probability* is accepted.

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30. Interview challenge: If you feel that you had a good first interview, what is the probability you will receive a second interview?

50% of all people who receive a first interview receive a second interview; 95% of your friends that got a second interview felt they had a good first interview; 75% of your friends that DID NOT get a second interview felt they had a good first interview. If you feel that you had a good first interview, what is the probability you will receive a second interview?

Answer

Supposing that my friends are a good representation of the entire population, the key to solving problems like this is to define the events carefully. Let's denote:

- pass: being invited to a second interview,
- fail: not being so invited,
- good: feel good about the first interview, and
- bad: don't feel good about the first interview.

So according to the problem, we need to find $p(\text{pass} | \text{good})$ given that:

$$\begin{aligned} p(\text{pass}) &= 0.5 \\ p(\text{good} | \text{pass}) &= 0.95 \\ p(\text{good} | \text{fail}) &= 0.75 \\ p(\text{pass})p(\text{good} | \text{pass}) &= 0.5 \times 0.95 = 0.475 \\ p(\text{fail})p(\text{good} | \text{fail}) &= 0.5 \times 0.75 = 0.375 \end{aligned}$$

Let's use the *Bayes theorem* such as:

$$\begin{aligned} p(\text{pass} | \text{good}) &= \frac{p(\text{good} | \text{pass}) \times p(\text{pass})}{p(\text{good})} \\ p(\text{good}) &= p(\text{good} | \text{pass}) \times p(\text{pass}) + p(\text{good} | \text{fail}) \times p(\text{fail}) \end{aligned}$$

where $p(\text{good})$ is given by:

$$\begin{aligned} p(\text{good}) &= p(\text{good} | \text{pass}) \times p(\text{pass}) + p(\text{good} | \text{fail}) \times p(\text{fail}) = 0.5(0.95 + 0.75) = 0.85 \\ p(\text{good}) &= 0.5(0.95 + 0.75) = 0.85 \end{aligned}$$

Thus

$$\begin{aligned} p(\text{pass} | \text{good}) &= \frac{0.95 \times 0.5}{0.85} \approx 0.559 \\ p(\text{pass} | \text{good}) &= 0.559 \end{aligned}$$

So feeling good about the interview only makes me slightly more likely to actually move on.

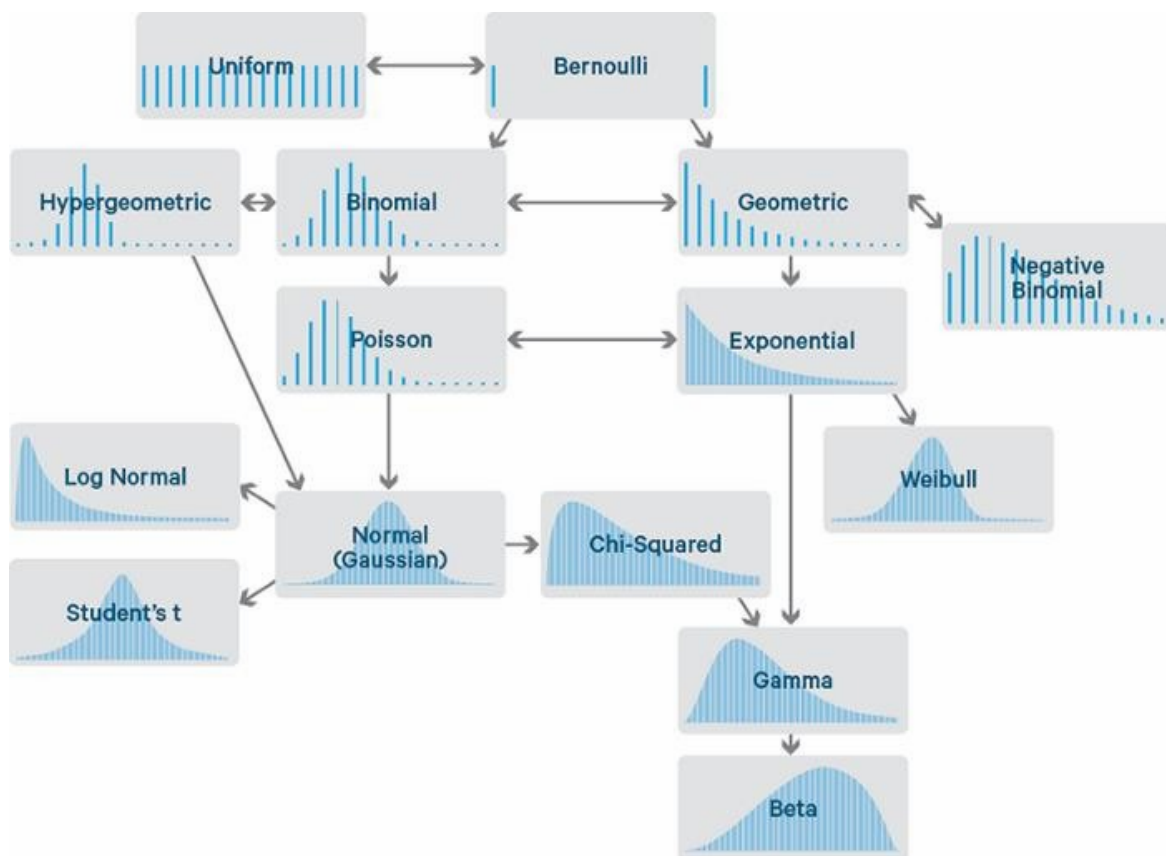
31. Name some *Probability Distributions* you know

There are two main types of probability distributions: *discrete* and *continuous*.

- **Discrete distributions**

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- *Bernoulli distribution*, which takes value 1 with probability p and value 0 with probability $q = 1 - p$, can be thought of as a model for the set of possible outcomes of any single experiment that asks a yes-no question.
- *Binomial distribution*, which describes the number of successes in a series of independent Yes/No experiments all with the same probability of success.
- *Poisson distribution*, which describes a very large number of individually unlikely events that happen in a certain time interval.
- **Continuous distributions:**
 - *Normal distribution*, it's one of the most relevant distributions, its importance is partly due to the central limit theorem which states that every variable can be modeled as a sum of many small independent, identically distributed variables with finite mean and variance approximately normal.
 - *Student's t-distribution*, useful for estimating unknown means of Gaussian populations
 - *Logistic distribution*, it's a symmetrical distribution, unimodal, and is similar in shape to the normal distribution.



32. Three Ants Challenge

Problem

Three ants are sitting at the corners of an equilateral triangle. Each ant randomly picks a direction and starts moving along the edge of the triangle. What is the probability that none of the ants collide?

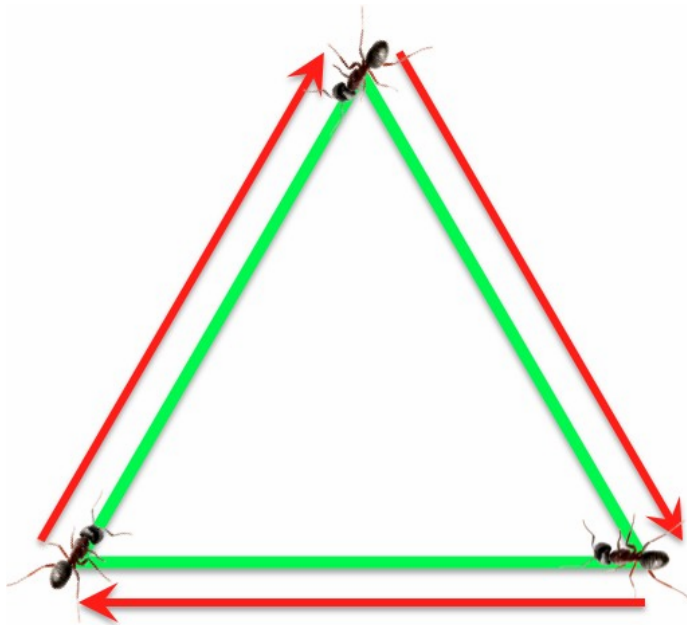
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Answer

Each ant has two possible ways to go: the edge on its left L and the edge on its right R . Now the only way no ant will collide is if they all walk in the same direction along the triangle (assuming they all move at the same speed). Overall the ways how the ants can move are:

1. LLL
2. LLR
3. LRL
4. RLL
5. LRR
6. RLR
7. RRL
8. RRR

We have a total of 8 ways how the 3 Ants can move, out of these, only RRR and LLL are the ways by which the Ants will never meet. So the probability of it is $2/8 = 0.25$



33. True Positive and False Positive challenge

A test has a true positive rate of 100% and a false positive rate of 5%. There is a population with a 1/1000 rate of having the condition the test identifies. Considering a positive test, what is the probability of having that condition?

Answer

Let's denote as A a person who has the disease, and let B a positive test. To calculate $P(A|B)$ and we will use the Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\neg A)P(\neg A)}$$

$$P(B|A)P(A)$$

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Where $P(B)$ is given by the law of total probability:

$$P(B) = P(B|A)P(A) + P(B|\text{not}A)P(\text{not}A)$$

A)

Now, we know that:

- Out of 1000 people, 1 person who has the disease will get true positive result, so $P(A) = 0.001$ and $P(\text{not}A) = 1 - P(A) = 0.999$.
- The test has a true positive rate of 100%, so $P(B|A) = 1$
- The test has a false positive rate of 5%, so $P(B|\text{not}A) = 0.05$.

Therefore,

$$P(A|B) = \frac{1 \cdot 0.001}{1 \cdot 0.001 + 0.05 \cdot 0.999} = \frac{0.001}{0.001 + 0.04995} = \frac{0.001}{0.05095} \approx 0.0198 \approx 0.02$$

Thus, there is only a 2% probability of one person having the disease even if the reports say that it has the disease.

34. What is the *Bayesian approach* to probability?

The **Bayesian approach** defines probability as the *measure of believability* one has about how a particular event occurs. It uses the **Bayes theorem** to help us update beliefs about random events once we've seen new evidence about the event, so we use Bayesian statistics to create different beliefs after new evidence is uncovered. It differs from *frequentist statistics* that rely only on data from repeated trials.

35. What's the difference between *Binomial Distribution* and *Geometric Distribution*?

The **Binomial distribution** describes the probability of obtaining k successes in n *Bernoulli experiments*, i.e an experiment which has only two possible outcomes, often call them *success* and *failure*. Its probability function describes the probability of getting exactly k successes in n independent *Bernoulli* trials:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

The **Geometric distribution** describes the probability of experiencing a *certain amount* of *failures* before experiencing the *first success* in a series of *Bernoulli experiments*. This probability is given by:

$$P(X=k) = p(1-p)^{k-1}$$

So as we can see, the key difference is that in a *binomial distribution*, there is a *fixed number of trials* meanwhile in a *geometric distribution*, we're interested in the *number of trials* required until we obtain a *success*.

36. What's the difference between *Cumulative Distribution Functions* and *Probability Density Functions*?

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The **Cumulative Distribution Function (CDF)** can be defined for any kind of random variable, i.e. *discrete* or *continuous*, and it tells us the probability that the random variable X takes a value less than or equal to a particular value x :

$$F(x) = \Pr[X \leq x]$$

The CDF is used to determine the probability that an observation will be greater than a certain value, or between two values.

A **Probability Density Function (PDF)** can be defined only for *continuous random variables* and it tells us the probability of the random variable X falling within a range of values (a, b) by computing the integral of this variable's PDF over that range:

$$F(x) = \Pr[a \leq X \leq b] = \int_a^b f_X(x) dx$$

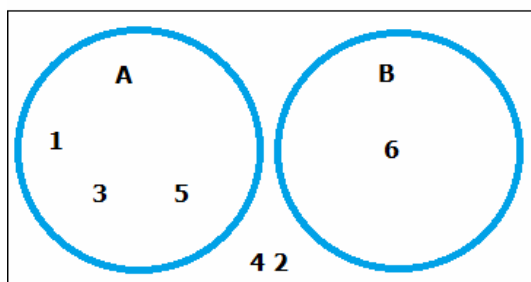
As we can see from the previous definition, the *PDF* is used when we want to know the probability that an observation relies on a certain range, with the restriction that the observation comes from a *continuous* random variable.

37. What's the difference between *Disjoint Events* and *Independent Events*?

- **Disjoint events** are events that *never occur at the same time*. These are also known as mutually exclusive events. These are often visually represented by a Venn diagram, such as the below. In this diagram, there is no overlap between event A and event B, so these two events never occur together.
- **Independent events** are unrelated events, i.e. an event A does not give any information about B and the outcome of one event does not impact the outcome of the other event. Independent events *can, and do often, occur together*.

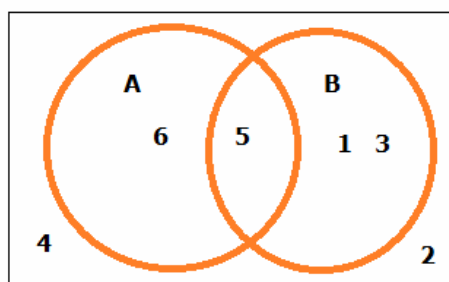
Disjoint Events

Event A: Get an odd Number
Event B: Get a 6



Overlapping Events

Event A: Get a number over 4
Event B: Get an odd number



38. What's the difference between the *likelihood* and the *posterior* probability in Bayesian statistics?

The *likelihood* and posterior probability are two quantities that are defined and used in the **Bayes Theorem**:

Data Science Learning Material

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

in which:

- $P(B|A)$ is the *likelihood*, this is the probability of observing the new evidence, given our initial hypothesis.
- $P(A|B)$ is the *posterior*; this is what we are trying to estimate.
- $P(A)$ and $P(B)$ is called the *prior* and the *marginal likelihood*, respectively.

An example case of use can be trying to find the probability that a given person has cancer. For that, we would initially just say it is whatever percent of the population has cancer. However, given additional evidence such as the fact that the person is a smoker, we can set the **likelihood** as the probability of being a smoker given that the person has cancer, and set as the **posterior** the probability of having cancer given that the person is a smoker.

Thus, given additional evidence such as the fact that the person is a smoker, we can update our probability, since the probability of having cancer is higher given that the person is a smoker. This allows us to utilize prior knowledge to improve our probability estimations.

39. When is an event *Independent of Itself*?

An event can only be *independent of itself* when either there is *no chance of it happening* or when it is *certain to happen*. To demonstrate this, let's remember that two events A and B are independent if:

$$P(A \cap B) = P(A)P(B)$$

Now if $B = A$, then

$$P(A \cap A) = P(A)P(A) = P(A)^2$$

and this result only can occur when $P(X) = 1$ or $P(X) = 0$ i.e. when:

- there's no probability of happening or
- it's 100% sure that it will happen.